

## Code-Layout, Readability and Reusability

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| <b>Date</b>          | 15 JULY 2026                                              |
| <b>Project Name</b>  | Smart Lender – AI Powered Loan Approval Prediction System |
| <b>Maximum Marks</b> | 5 Marks                                                   |

### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks                                                                           |
|------|---------------------------|-----------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout          | Yes                           | Proper indentation followed in Python and Flask files.                            |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes                           | Project separated into dataset, model, templates, static, and application files.  |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes                           | Variables like loan_amount, credit_history, and applicant_income are descriptive. |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes                           | Functions clearly indicate their purpose, such as prediction and preprocessing.   |
| 5    | Code Comments             | Inline and block comments explain logic       | Partial                       | Comments are included for important sections but can be improved further.         |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes                           | Separate modules used for preprocessing, model training, and prediction.          |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes                           | Repeated code has been minimized for better maintainability.                      |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Partial                       | Basic exception handling is implemented for user input and prediction errors.     |

### Reusable Components / Modules:

| S.No | Component / Module Name   | Language / Technology | Where Reused                | Reusability Level (High / Medium / Low) |
|------|---------------------------|-----------------------|-----------------------------|-----------------------------------------|
| 1    | Data Preprocessing Module | Python, Pandas        | Model Training & Prediction | High                                    |
| 2    | Machine Learning Models   | Scikit-learn, XGBoost | Loan Prediction             | High                                    |
| 3    | Prediction Module         | Flask, Python         | Web Application             | High                                    |
| 4    | User Interface            | HTML, CSS, Bootstrap  | Loan Prediction Form        | Medium                                  |
| 5    | Model Loading Module      | Pickle, Flask         | Real-time Prediction        | High                                    |

**Overall Code Quality Assessment:**

| Aspect                   | Rating (1-5) | Comments                                                                          |
|--------------------------|--------------|-----------------------------------------------------------------------------------|
| Code Layout & Structure  | 5            | Well-organized project structure with separate modules.                           |
| Readability              | 5            | Meaningful naming conventions and clean code improve readability.                 |
| Reusability              | 5            | Reusable modules simplify maintenance and future enhancements.                    |
| Documentation / Comments | 4            | Adequate comments provided; additional documentation can further improve clarity. |
| Overall Score            | 4.8/ 5       | The codebase is modular, maintainable, and follows good coding practices.         |